

Purpose

The purpose of this project is to explore what factors influence customer conversion in digital marketing campaigns using an interactive Streamlit dashboard. The dashboard combines engagement analysis, statistical testing, ROAS/CAC analysis, funnel analysis, and customer segmentation to better understand customer behavior and campaign performance.

This project also explores how interactive visualization, machine learning, and dynamic findings can be combined into one dashboard to support marketing analysis and decision-making.

Findings & Outlook

Key Findings (Dynamic Data)

This project explores customer conversion behavior in digital marketing campaigns through an interactive Streamlit dashboard combining engagement analysis, statistical testing, ROAS/CAC evaluation, and customer segmentation.

First, engagement showed a positive relationship with conversion. Among the filtered 8,000 users, time on site had the strongest correlation with conversion ($r = 0.130$). Users with above-average time on site converted at 91.3%, compared to 84.0% for users below the median. This suggests that users with higher engagement are more likely to convert.

Second, channel performance differences were not statistically significant ($\chi^2 = 2.78$, $p = 0.5944$). Referral had the highest conversion rate (88.3%), while Social Media had the lowest (86.8%), with a difference of 1.5 percentage points between them.

Third, ROAS and CAC analysis identified PPC as the most efficient channel under the assumed revenue model of USD 150 per conversion. PPC had the highest ROAS ($0.03\times$) and the lowest customer acquisition cost at approximately USD 5,612 per converting user.

Finally, K-Means segmentation identified clear differences between customer groups. The High-value segment (3,435 users) converted at 100.0%, while the At-risk segment (988 users) converted at 0.0%. Funnel analysis also showed that the largest drop-off happened between the Opened Email and Clicked Email stages.

Next Steps

Based on these findings, future improvements could focus on increasing user engagement through better navigation, stronger calls-to-action, and more personalized content.

Future analysis could also focus on improving budget allocation by combining ROAS, CAC, and retention metrics to compare channel performance more effectively.

In addition, segment-specific campaigns could be developed for At-risk users through targeted email campaigns and re-engagement strategies.

If given additional time and resources, this project could be expanded using real-world marketing data from companies such as Amazon or Target. Access to larger datasets would allow for more accurate analysis of customer engagement, conversion behavior, ROAS, CAC, and customer retention across different marketing channels. Future versions could also include real-time campaign tracking, recommendation systems based on customer segments, predictive machine learning models, and automated A/B testing systems.

Collaboration Statement

This project was completed independently with the assistance of AI tools including ChatGPT and Claude.

AI tools were used to assist with debugging Streamlit and Plotly code, refining dashboard styling and layout, brainstorming visualization ideas, improving narrative explanations, and organizing the project structure.

One of the major improvements in the final version was the addition of dynamically generated findings and outlook sections. Instead of static written summaries, the dashboard recalculates correlations, conversion gaps, ROAS, CAC, funnel drop-off rates, and segmentation statistics based on the selected filters.

Claude AI was used to assist with implementing parts of this dynamic narrative generation logic, including integrating calculated metrics into Streamlit markdown components using Python f-strings. This changed the dashboard from a static visualization project into a more interactive analytical storytelling system.

The initial version of the dashboard focused mainly on three visualizations: engagement vs conversion, channel ROI, and funnel analysis. With assistance from Claude AI for statistical analysis logic and dashboard features, the final version expanded to include channel drill-down analysis, A/B significance testing, ROAS & CAC evaluation, K-Means customer segmentation, radar charts, and dynamic insight generation.

Initially, the dashboard used raw TimeOnSite values directly in the visualization. After reviewing the first chart, it became difficult to identify broader engagement trends because the data was too granular. Following GPT's suggestion, the final version grouped users into pandas-generated intervals (`pd.qcut`) and calculated the average conversion rate for each group. This made the relationship between engagement and conversion behavior clearer.

K-Means clustering was also a new concept and visualization approach learned during the development process with assistance from Claude AI. Claude helped explain how behavioral segmentation works and how clustering could be used to identify customer groups based on engagement patterns.